DPUSGEScreenServer

Sends the dynamic contents of a window over the network. Implements the ScreenSend protocol, which can be received using *DPUSGETextureScreen*.

1. Static parameters:

Source	DPU instance name window whose contents are sent. Instances of DPUSGEView, DPUMMI and DPUBirdsEye can be used. An empty string captures the contents of the desktop, i.e. any window that is located at the given coordinates. Default: *SGEView*
Password	Login password for this ScreenSend server. If one is entered, the same password must be set in the client. Default: (*empty*)
Port	UDP port of the ScreenSend server. Default: 5955
PacketSize	Size of the network packets that are sent to the client. The default value (1452) is optimized for standard LAN networks. If the network supports Jumbo Frames, larger val-ues can be used.
X, Y	Position of the top-left edge of the transmitted area within the selected window (or on the desktop).
W, H	Size of the transmitted area within the selected window (or on the desktop). **Note**: There is no default value; the parameters must be set.
Subsample	Proper divisor of the frame rate. The following equation is used: Transmission frame rate = Source window update rate / *Subsample*. Default: 1
Refresh	Screen refresh rate. Every *Refresh'th* image is transmitted completely, even if the contents have not changed. This mechanism safeguards against packet loss. Default: 10, but in a LAN 100 can be used as well (which reduces the data rate).

2. Outputs:

FrameNumber	Frame number of the last sent image.

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3. Functional principle:

Using *DPUSGEScreenServer*, the contents of a graphics window are captured. Using the ScreenSend protocol, they are then sent using a UDP connection over the network.

4. Limitations:

Currently, only a single client (such as *DPUSGETextureScreen*) can be connected to a *DPUSGEScreen-Server* instance.

The data is sent uncompressed. If a SGEView window is sent (whose contents usu-ally change constantly) then the amount of data can quickly exceed the available network bandwidth.

For example, when using a standard Gigabit connection that is used for general networking over a switch, not more than 640 x 480 pixels can be transmitted at an update rate of 60 Hz.

If a dedicated Gigabit connection is used (which connects the sender and receiver directly), the window size can be at most 960 x 600 pixels at 60 Hz.

Larger views require expensive 10 GE connections, as well as significant amounts of CPU time.

Captures from the desktop are slower than direct window captures and are also prone to graphics artifacts. If possible, direct window captures (where Source is not empty) should be used.

5. Dependencies to other DPUs:

The DPU that provides the captured window must run on the computer on which **DPUSGEScreen-Server** is executed. The W and H parameters must not be larger than the window size of the captured window.

6. Example:

This DPU works well using the following script:

```
DPUSGEScreenServer ScreenServer
{
      Computer = {VIS};
      Index = 62;

      Source = SGEView;
      Subsample = 1;

      W = 60;
      H = 480;
      Refresh = 100;
};
```